GATE CS — Compiler Design

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
Parsing (LL/LR, FIRST/FOLLOW)	25	38	2
Intermediate code, optimisation, data-flow, runtime	15	24	2
Syntax-directed translation	10	18	2
Lexical analysis, symbol table, phases	7	8	2
Total	**57**	**88**	**8**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

Parsing (LL/LR, FIRST/FOLLOW)

FIRST/FOLLOW, LL(1), LR(0)/SLR/LALR item sets, conflicts. · 25 questions, 38 marks over 15 papers (≈2.5 marks/paper)

A. Past-paper questions (25)

2026 CS1 · Q.28 · 1 mark

Which of the following statements is/are true?
(A) LL(1) parser uses backtracking
(B) For a grammar to be LL(1), it must be left-recursive
(C) For a grammar to be LL(1), it must be left-factored
(D) The LL(1) parsers are more powerful than the SLR parsers

2026 CS2 · Q.41 · 2 marks

Consider the canonical LR(0) parsing of the grammar below using terminals {a, b, c} and non-terminals {A, B, C, S} with S as the start symbol. S → ACB A → aA | ε C → cC | ε B → bB | b Which one of the following options gives the number of shift-reduce conflicts that will occur in the LR(0) ACTION table?
(A) 2
(B) 3
(C) 4
(D) 5

2025 CS1 · Q.46 · 2 marks

Which of the following statement(s) is/are TRUE while computing First and Follow during top down parsing by a compiler?
(A) For a production A → ε, ε will be added to First(A).
(B) If there is any input right end marker, it will be added to First(S), where S is the start symbol.
(C) For a production A → ε, ε will be added to Follow(A).
(D) If there is any input right end marker, it will be added to Follow(S), where S is the start symbol.

2025 CS2 · Q.40 · 2 marks

Given a Context-Free Grammar G as follows: S → Aa | bAc | dc | bda A→d Which ONE of the following statements is TRUE?
(A) G is neither LALR(1) nor SLR(1)
(B) G is CLR(1), not LALR(1)
(C) G is LALR(1), not SLR(1)
(D) G is LALR(1), also SLR(1)

2025 CS2 · Q.51 · 2 marks
Consider two grammars G1 and G2 with the production rules given below: G1 : S → if E then S | if E then S else S | a E→b G2 : S → if E then S | M M → if E then M else S | c E→b where if, then, else, a, b, c are the terminals. Which of the following option(s) is/are CORRECT?
(A) G1 is not LL(1) and G2 is LL(1).
(B) G1 is LL(1) and G2 is not LL(1).
(C) G1 and G2 are not LL(1).
(D) G1 and G2 are ambiguous.

2024 CS1 · Q.26 · 1 mark
Which of the following is/are Bottom-Up Parser(s)?
(A) Shift-reduce Parser
(B) Predictive Parser
(C) LL(1) Parser
(D) LR Parser

2024 CS1 · Q.33 · 1 mark
Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below. Operator Precedence Associativity + Highest Left − High Right ∗ Medium Right / Low Right The value of the expression 3 + 1 + 5 ∗ 2 / 7 + 2 − 4 − 7 − 6 / 2 as per the above rules is _____

2024 CS1 · Q.38 · 2 marks
Consider the following grammar G, with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3). S → daT | (1) T → aS | bT | (2) R → (3) | ε The set of terminals is {a, b, c, d, f}. The FIRST and FOLLOW sets of the different non-terminals are as follows. FIRST(S) = {c, d, f}, FIRST(T) = {a, b, ε}, FIRST(R) = {c, ε} FOLLOW(S) = FOLLOW(T) = {c, f, $}, FOLLOW(R) = {f} Which one of the following options CORRECTLY fills in the incomplete productions?
(A) (1) S → Rf (2) T → ε (3) R → cTR
(B) (1) S → fR (2) T → ε (3) R → cTR
(C) (1) S → fR (2) T → cT (3) R → cR
(D) (1) S → Rf (2) T → cT (3) R → cR

2024 CS2 · Q.40 · 2 marks
Consider the following context-free grammar where the start symbol is S and the set of terminals is {a,b,c,d}. S → AaAb|BbBa A → cS|ε B → dS|ε The following is a partially-filled LL(1) parsing table. a b c d $ S S → AaAb S → BbBa (1) (2) A A→ε (3) A → cS B (4) B→ε B → dS Which one of the following options represents the CORRECT combination for the numbered cells in the parsing table? Note: In the options, “blank” denotes that the corresponding cell is empty.
(A) (1) S → AaAb (2) S → BbBa (3) A → ε (4) B → ε
(B) (1) S → BbBa (2) S → AaAb (3) A → ε (4) B → ε
(C) (1) S → AaAb (2) S → BbBa (3) blank (4) blank
(D) (1) S → BbBa (2) S → AaAb (3) blank (4) blank

2024 CS2 · Q.65 · 2 marks
Consider the following augmented grammar, which is to be parsed with a SLR parser. The set of terminals is {a,b,c,d,#,@} S′ → S S → SS|Aa|bAc|Bc|bBa A → d# B→@ Let I0 = CLOSURE({S ′ →• S}). The number of items in the set GOTO(I0 , S) is _____

2022 · Q.13 · 1 mark
Which one of the following statements is TRUE?
(A) The LALR (1) parser for a grammar G cannot have reduce-reduce conflict if the LR (1) parser for G does not have reduce-reduce conflict.
(B) Symbol table is accessed only during the lexical analysis phase.
(C) Data flow analysis is necessary for run-time memory management.
(D) LR (1) parsing is sufficient for deterministic context-free languages.

2022 · Q.29 · 1 mark
Consider the augmented grammar with {, *, (,), id } as the set of terminals. S S S S R|R R R*P | P P (S) | id If I 0 is the set of two LR (0) items {[S S•], [S S• R]} , then goto(closure(I0),) contains exactly _____ items.

2021 · Q.5 · 1 mark [OCR]
Consider the following statements. Si: Every SLR(1) grammar is unambiguous but there are certain unambiguous grammars that are not SLR(1). So: For any context-free grammar, there is a parser that takes at most O(n*) time to parse a string of length n. Which one of the following options is correct?
(A) | Si is true and S2 is false
(B) | S; is false and Sy is true
(C) | S; is true and S¢ is true
(D) | S; is false and Sy is false

2021 · Q.31 · 2 marks [OCR]
Consider the following context-free grammar where the set of terminals is {a,b, c,d, f}. S+daT|Rf T-aS|baT |e R-caTRle The following is a partially-filled LL(1) parsing table. a b e d f $ s @ S—+daT Q T T+aS T+baT (3) Toe @ R R-caTR R-e Which one of the following choices represents the correct combination for the num- bered cells in the parsing table (“blank” denotes that the corresponding cell is empty)? (A)|@ssrRf Q@sSsRf @BTs+e @Te (B)|@blnk Q@SsRf @Tse @T-e (C)|@ss+rf @pblank G@)dlank (GT. (D)|@ blank @)S+RFf @)bdlank @) blank CS, Set-1 - Copyright © GATE 2021

2020 · Q.24 · 1 mark [OCR]
Consider the following grammar. S—>aSB\d Bob The number of reduction steps taken by a bottom-up parser while accepting the string aaadbbb is

2019 · Q.3 · 1 mark [OCR]
Which one of the following kinds of derivation is used by LR parsers?
(A) Leftmost
(B) Leftmost in reverse
(C) Rightmost
(D) Rightmost in reverse

2019 · Q.19 · 1 mark [OCR]
Consider the grammar given below: S— Aa A—BD B-ble D—dle Let a, b, d, and $ be indexed as follows: a |b |d |$ 3 |2 {1 40 Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, $}, then the answer should be 3210) Answer: cs 41s GATE 2019 Computer Science and Information Technology Set-2

2019 · Q.43 · 2 marks [OCR]
Consider the augmented grammar given below: S'S S— (L) | id L—>LS|S Let Ip = CLOSURE ({[S' — *S]}). The number of items in the set GOTO (Ib , ( ) is:

2018 · Q.8 · 1 mark
Which one of the following statements is FALSE?
(A) Context-free grammar can be used to specify both lexical and syntax rules.
(B) Type checking is done before parsing.
(C) High-level language programs can be translated to different Intermediate Representations.
(D) Arguments to a function can be passed using the program stack. CS 2/20 GATE 2018

2018 · Q.38 · 2 marks
Consider the following parse tree for the expression a#bcd#e#f, involving two binary operators $ and #. Which one of the following is correct for the given parse tree?
(A) $ has higher precedence and is left associative; # is right associative
(B) # has higher precedence and is left associative; $ is right associative
(C) $ has higher precedence and is left associative; # is left associative
(D) # has higher precedence and is right associative; $ is left associative CS 12/20 GATE 2018

2017 CS1 · Q.17 · 1 mark [OCR]
Consider the following grammar: P—xORS Q—> yz |z Rowle S—>y What is FOLLOW(Q)?
(A) {R}
(B) tw} (©) tw x}
(D) fw, $}

2017 CS1 · Q.43 · 2 marks [OCR]
Consider the following grammar: stmt -> if expr then expr else expr; stmt | 0 expr -> term relop term | term term -> id | number id eae. |. bY te number -> [0-9] where relop is a relational operator (e.g..<.>....). O refers to the empty statement, and if, then, else are terminals. Consider a program P following the above grammar containing ten if terminals. The number of control flow paths in P is . For example, the program if e, then e: else e; has 2 control flow paths. ¢. > e2 and e: > 63. =====PAGE 20=====

2017 CS2 · Q.6 · 1 mark [OCR]
Which of the following statements about parser is/are CORRECT? I. Canonical LR is more powerful than SLR. II. SLR is more powerful than LALR. III. SLR is more powerful than Canonical LR.
(A) Lonly
(B) I only
(C) II only
(D) I and III only =====PAGE 4=====

2017 CS2 · Q.32 · 2 marks [OCR]
Consider the following expression grammar G : E ->E — T ¥Se T —-> fT 7S F F > (F)Svid Which of the following grammars is not left recursive. but is equivalent to G? (QE=>E-T|T (B) E -> TE’ To T+E|E E’ -> -TE’|e F -> (E) | id To T+FEIE F -> (BE) | id (G B => tx (D) E -> TX | (TX) X -> -TX |e X -> -TX | +TX |e T <> FY T -> id YS ao: EE |€ EB aoa CB) [. ae =====PAGE 13=====

2016 · Q.45 · 2 marks
The attributes of three arithmetic operators in some programming language are given below. Operator Precedence Associativity Arity + High Left Binary − Medium Right Binary ∗ Low Left Binary The value of the expression 2 − 5 + 1 − 7 ∗ 3 in this language is .

B. New practice questions with solutions

1. (1 mark · NAT) For the grammar

E → T E′, E′ → + T E′ | ε, T → F T′, T′ → * F T′ | ε, F → (E) | id

the number of terminals in FOLLOW(F) is _____.

> **4.** FOLLOW(F) includes FIRST(T′) minus ε, which is *{}. Because T′ can be empty it also includes FOLLOW(T) = {+,), $}. The set is {*, +,), $}.

2. (2 marks · NAT) For the grammar S → C C, C → c C | d, let x be the number of states in the canonical LR(1) automaton and y the number in the LR(0) automaton. The value of x − y is _____.

> **3.** The LR(1) automaton has 10 states and the LR(0) automaton has 7. The extra 3 come from states that share an LR(0) core but have different lookaheads.

Intermediate code, optimisation, data-flow, runtime

Three-address code, basic blocks, liveness/data-flow, optimisation, runtime environment. · 15 questions, 24 marks over 15 papers (≈1.6 marks/paper)

A. Past-paper questions (15)

2026 CS1 · Q.42 · 2 marks

Consider the control flow graph shown in the figure. Which one of the following options correctly lists the set of redundant expressions (common subexpressions) in the basic blocks B4 and B5? Note: All the variables are integers.
(A) B4: { b + i } B5: { c + m }
(B) B4: { g ∗ k } B5: { c + m }
(C) B4: { g ∗ k, b + i } B5: { }
(D) B4: { g ∗ k } B5: { }

2026 CS2 · Q.17 · 1 mark

In C runtime environment, which one of the following is stored in heap?
(A) A static variable declared inside a function
(B) An array of integers declared inside a function
(C) A dynamically allocated array of integers created using malloc() function call
(D) Return address of a function

2026 CS2 · Q.45 · 2 marks

Consider the control flow graph given below. Which one of the following options is the set of live variables at the exit point of each basic block?
(A) B1:{a, b, c, e, f}, B2:{d, e}, B3:{b, c, e, f}, B4:∅
(B) B1:∅, B2:{d, e}, B3:{a, c, f}, B4:∅
(C) B1:{a, b, c, e, f}, B2:{d, e}, B3:{c, e, f}, B4:∅
(D) B1:∅, B2:{d, e, f}, B3:{a, b, c, e, f}, B4:∅

2025 CS1 · Q.13 · 1 mark

Which ONE of the following techniques used in compiler code optimization uses live variable analysis?
(A) Run-time function call management
(B) Register assignment to variables
(C) Strength reduction
(D) Constant folding

2025 CS1 · Q.52 · 2 marks

Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____. (Answer in integer) 1001: i = 1 1002: j = 1 1003: t1 = 10*i 1004: t2 = t1+j 1005: t3 = 8*t2 1006: t4 = t3-88 1007: a[t4] = 0.0 1008: j = j+1 1009: if j <= 10 goto 1003 1010: i = i+1 1011: if i <= 10 goto 1002 1012: i = 1 1013: t5 = i-1 1014: t6 = 88*t5 1015: a[t6] = 1.0 1016: i = i+1 1017: if i <= 10 goto 1013

2025 CS2 · Q.21 · 1 mark

Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Backpatching can be used to generate code for Boolean expression in one pass. (II) Backpatching can be used to generate code for flow-of-control statements in one pass. Which ONE of the following options is CORRECT?

(A) Only (I) is correct.

(B) Only (II) is correct.

(C) Both (I) and (II) are correct.

(D) Neither (I) nor (II) is correct.

2024 CS1 · Q.39 · 2 marks

Consider the following pseudo-code. L1: t1 = −1 L2: t2 = 0 L3: t3 = 0 L4: t4 = 4 ∗ t3 L5: t5 = 4 ∗ t2 L6: t6 = t5 ∗ M L7: t7 = t4 + t6 L8: t8 = a[t7] L9: if t8 <= max goto L11 L10: t1 = t8 L11: t3 = t3 + 1 L12: if t3 < M goto L4 L13: t2 = t2 + 1 L14: if t2 < N goto L3 L15: max = t1 Which one of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively ?

(A) 6 and 6

(B) 6 and 7

(C) 7 and 7

(D) 7 and 6

2024 CS2 · Q.43 · 2 marks

Consider the following expression: x[i] = (p + r) ∗ −s[i] + u/w. The following sequence shows the list of triples representing the given expression, with entries missing for triples (1), (3), and (6). (0) + p r (1) (2) uminus (1) (3) (4) / u w (5) + (3) (4) (6) (7) = (6) (5) Which one of the following options fills in the missing entries CORRECTLY?

(A) (1) =[]si (3) * (0) (2) (6) []=xi

(B) (1) []=si (3) –(0)(2) (6) =[] x(5)

(C) (1) =[] si (3) * (0)(2) (6) []= x(5)

(D) (1) []= si (3) –(0)(2) (6) =[] xi

2023 · Q.11 · 1 mark

Consider the following statements regarding the front-end and back-end of a compiler. S1: The front-end includes phases that are independent of the target hardware. S2: The back-end includes phases that are specific to the target hardware. S3: The back-end includes phases that are specific to the programming language used in the source code. Identify the CORRECT option.

(A) Only S1 is TRUE.

(B) Only S1 and S2 are TRUE.

(C) S1, S2, and S3 are all TRUE.

(D) Only S1 and S3 are TRUE. CS

2023 · Q.37 · 2 marks
Consider the control flow graph shown. ENTRY B1 i = m − 1 j = n a = 10 B2 i = i + 1 j = j − 1 B3 a = 20 B4 i = a + 1 EXIT Which one of the following choices correctly lists the set of live variables at the exit point of each basic block?
(A) B1: {}, B2: {a}, B3: {a}, B4: {a}
(B) B1: {i, j}, B2: {a}, B3: {a}, B4: {i}
(C) B1: {a, i, j}, B2: {a, i, j}, B3: {a, i}, B4: {a}
(D) B1: {a, i, j}, B2: {a, j}, B3: {a, j}, B4: {a, i, j} CS

2021 · Q.50 · 2 marks [OCR]
Consider the following C code segment: a=b+tc; e=art+til; d=b+t+ec; f=d+i; g=erf,; In a compiler, this code segment is represented internally as a directed acyclic graph (DAG). The number of nodes in the DAG is CS, Set-1 - Copyright © GATE 2021

2017 CS1 · Q.12 · 1 mark [OCR]
Consider the following intermediate program in three address code p=a-b q=p*e p=uty Gp pest sq Which one of the following corresponds to a static single assignment form of the above code?
(A) Bj =a=b
(B) Pp; = a%ob 4, =P,* c¢ d4 @Ps* ¢ Pig = et ow. DR, =a * v GQ = Pit GY WF Pat Ay
(C) pj var- b () Py, = an-b Gi = P55 se Go= Dp ® Pp; =u*v P> =~u*v as spy *F a5 Gq, =ptq =====PAGE 8=====

2017 CS1 · Q.52 · 2 marks [OCR]
Consider the expression (a—1)*(((b+ce)/3)+d)). Let X be the minimum number of registers required by an optimal code generation (without any register spill) algorithm for a load/store architecture. in which (i) onlv load and store instruetions can have memory operands and (ii) arithmetic instructions can have only register or immediate operands. The value of X is

2016 · Q.19 · 1 mark
Consider the following code segment. x = u - t; y = x * v; x = y + w; y = t - z; y = x * y; The minimum number of total variables required to convert the above code segment to static single assignment form is .

2016 · Q.36 · 2 marks
What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed? a=3; void n(x) {x = x * a; print(x);} void m(y) {a = 1; a = y - a; n(a); print(a);} void main() {m(a);}
(A) 6, 2
(B) 6, 6
(C) 4, 2
(D) 4, 4

B. New practice questions with solutions

1. (1 mark · NAT) How many basic blocks does this three-address code contain?

```
1: i = 0
2: s = 0
3: if i >= 10 goto 8
```

```
4: t = i * 2
5: s = s + t
6: i = i + 1
7: goto 3
8: print s
```

> **4.** The leaders are instruction 1 (first), 3 (target of the goto at 7), 4 (follows a conditional jump) and 8 (target of the jump at 3). The blocks are 1–2, 3, 4–7 and 8.

2. (2 marks · NAT) A control flow graph has these blocks. B1: a = 1; b = 2. B2: c = a + b; if c > 5 goto B4. B3: d = c * 2; goto B5. B4: d = a. B5: print d. B2 falls through to B3. The number of variables live at the exit of B2 is _____.

> **2.** B3 uses c (before defining d) and B4 uses a. So live-out of B2 is {a, c}. Variable b is not used after B2, and d is defined before it is used in both successors.

Syntax-directed translation

Synthesized/inherited attributes, SDT evaluation. · 10 questions, 18 marks over 15 papers (≈1.2 marks/paper)

A. Past-paper questions (10)

2026 CS1 · Q.53 · 2 marks

Consider the following two syntax-directed definitions SDD1 and SDD2 for type declarations. SDD1 SDD2 Grammar Semantic Rules Grammar Semantic Rules (G1) (G2) D →TV D. type = T. type D → D1 id D. type = D1 . type V. type = T. type put(id. entry, D1 . type) T → int T. type = int D → T id D. type = T. type T → float T. type = float put(id. entry, T. type) V → V1 id V1 . type = V. type put(id. entry, V. type) T → int T. type = int T → float T. type = float V → id put(id. entry, V. type) D is the start symbol, and int, float and id are the three terminals. The non-terminal V1 is the same as V and the non-terminal D1 is the same as D. Here, the subscript is used to differentiate the grammar symbols on the two sides of a production. The function put updates the symbol table with the type information for an identifier. Let P and Q be the languages specified by grammars G1 and G2, respectively. Which of the following statements is/are true?
(A) The languages P and Q are the same
(B) SDD2 is S-attributed and contains only synthesized attributes
(C) SDD1 is L-attributed and contains only inherited attributes
(D) The specifications of SDD1 and SDD2 are such that the same entries get added to the symbol table

2025 CS2 · Q.22 · 1 mark

Given the following syntax directed translation rules: Rule 1: R → AB {B. i = R. i − 1; A. i = B. i; R. i = A. i + 1; } Rule 2: P → CD {P. i = C. i + D. i; D. i = C. i + 2; } Rule 3: Q → EF {Q. i = E. i + F. i; } Which ONE is the CORRECT option among the following?
(A) Rule 1 is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
(B) Rule 1 is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L- attributed; Rule 3 is S-attributed and L-attributed
(C) Rule 1 is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed
(D) Rule 1 is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed

2024 CS1 · Q.37 · 2 marks

Consider the following syntax-directed definition (SDD). S → DHTU { S. val = D. val + H. val + T. val + U. val; } D → "M"D1 { D. val = 5 + D1 . val; } D→ε { D. val = −5; } H → "L"H1 { H. val = 5 ∗ 10 + H1 . val; } H→ε { H. val = −10; } T → "C"T1 { T. val = 5 ∗ 100 + T1 . val; } T→ε { T. val = −5; } U → "K" { U. val = 5; } Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute S. val)?
(A) 45
(B) 50
(C) 55
(D) 65

2024 CS2 · Q.29 · 1 mark

Which of the following statements is/are FALSE? An attribute grammar is a syntax-directed definition (SDD) in which the functions
(A) in the semantic rules have no side effects The attributes in a L-attributed definition cannot always be evaluated in a depth-
(B) first order
(C) Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed All L-attributed definitions based on LR(1) grammar can be evaluated using a
(D) bottom-up parsing strategy

2023 · Q.60 · 2 marks

Consider the syntax directed translation given by the following grammar and se- mantic rules. Here N , I, F and B are non-terminals. N is the starting non-terminal, and #, 0 and 1 are lexical tokens corresponding to input letters “#”, “0” and “1”, respectively. X.val denotes the synthesized attribute (a numeric value) associated with a non-terminal X. I1 and F1 denote occurrences of I and F on the right hand side of a production, respectively. For the tokens 0 and 1, 0.val = 0 and 1.val = 1. N → I #F N.val = I.val + F.val I → I1 B I.val = (2 I1 .val) + B.val I → B I.val = B.val F → B F1 F.val = 12 (B.val + F1 .val) F → B F.val = 12 B.val B → 0 B.val = 0.val B → 1 B.val = 1.val The value computed by the translation scheme for the input string 10#011 is . (Rounded off to three decimal places) CS

2022 · Q.65 · 2 marks

Consider the following grammar along with translation rules. S S1 # T {S•val S1•val * T•val } S T {S• val T• val } T T1 % R {T• val T1•val R• val } T R {T•val R•val } R id {R•val id•val } Here # and % are operators and id is a token that represents an integer and id•val represents the corresponding integer value. The set of non-terminals is {S , T , R , P} and a subscripted non-terminal indicates an instance of the non-terminal. Using this translation scheme, the computed value of S•val for root of the parse tree for the expression 20 #10%5 #8%2%2 is _____.

2021 · Q.26 · 2 marks [OCR]

Consider the following grammar (that admits a series of declarations, followed by expressions) and the associated syntax directed translation (SDT) actions, given as pseudo-code: P —+ D* E D -— int ID {record that ID.lexeme is of type int} D -+ bool ID {record that ID.lexeme is of type bool} E + £&,+ Ey, {check that E,.type = Ey.type = int; set E.type := int} E - !E, {check that E,.type = bool; set E.type := bool} E - ID {set E.type := int} With respect to the above grammar, which one of the following choices is correct?
(A) | The actions can be used to correctly type-check any syntactically correct program.
(B) | The actions can be used to type-check syntactically correct integer variable declarations and integer expressions.
(C) | The actions can be used to type-check syntactically correct boolean variable declarations and boolean expressions.
(D) | The actions will lead to an infinite loop. CS, Set-1 - Copyright © GATE 2021

2020 · Q.33 · 2 marks [OCR]
Consider the productions A > PQ and A> XY. Each of the five non-terminals A, P. Q. X, and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules. Rule 1: P.i=A.i+ 2, Qi=Pit+A.i, and A.s=P.s+Q.s Rule 2: X.i= A.i+ Y.s and Y.i= X.s + Ai Which one of the following is TRUE? (a) Both Rule 1 and Rule 2 are L-attributed. (8) Only Rule 1 is L-attributed. () Only Rule 2 is L-attributed. (0) Neither Rule 1 nor Rule 2 is L-attributed.

2019 · Q.36 · 2 marks [OCR]
Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X,, Xz, X3, X4, X5, and Xg be the placeholders for the non-terminals D, T, L or Li in the following table: Production rule Semantic action D-TL X,.type = X2.type T— int T.type = int T = float T.type = float LL, id X3.type = X4.type add'lype(id.entry, X5.type) Lid addType(id.entry, X¢.type) Which one of the following are the appropriate choices for X,, X2, X3 and X4?
(A) X, =L ,X2. =T, X3 =14,X,=L
(B) X, =T ,X2 =L, X3 = 14, X4 =T (©) X1 = L, Xp = L, X3 = 11, X, =T
(D) X, =T X= L, X3 =T,Xq = Ly

2016 · Q.46 · 2 marks
Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals {S, A} and terminals {a, b}. S −→ aA { print 1 } S −→ a { print 2 } A −→ Sb { print 3 } Using the above SDTS, the output printed by a bottom-up parser, for the input aab is:
(A) 1 3 2
(B) 2 2 3
(C) 2 3 1
(D) syntax error

B. New practice questions with solutions

1. (1 mark · MCQ) Consider E → E_1 + T {E.val = E_1.val + T.val}, E → T {E.val = T.val}, T → num {T.val = num.lexval}. The attribute val is

A. inherited, and the definition is L-attributed only
B. synthesized, and the definition is S-attributed
C. inherited, and the definition is S-attributed
D. neither synthesized nor inherited

(B). Each val is computed from attributes of the children, so it is synthesized, and a definition using only synthesized attributes is S-attributed.

2. (2 marks · NAT) The grammar E → E + T | T, T → T * F | F, F → id has a semantic action on every production, run when that production is reduced. A bottom-up parser parses id + id * id. The number of semantic actions executed is _____.

8. The rightmost derivation in reverse uses F → id three times, T → F twice, T → T * F once, E → T once and E → E + T once. 3 + 2 + 1 + 1 + 1 = 8.

Lexical analysis, symbol table, phases

Tokens and lexical patterns, symbol table, compiler phases. · 7 questions, 8 marks over 15 papers (≈0.5 marks/paper)

A. Past-paper questions (7)

2026 CS2 · Q.35 · 1 mark
A lexical analyzer uses the following token definitions letter → [A − Za − z] digit → [0 − 9] id → letter (letter | digit)* number → digit + ws → (blank | tab | newline)+ For the string given below, x1 23mm 78 y 7z zz5 14A 8H AaYcD the number of tokens (excluding ws) that will be produced by the lexical analyzer is _____. (answer in integer)

2025 CS1 · Q.12 · 1 mark
Which ONE of the following statements is FALSE regarding the symbol table?
(A) Symbol table is responsible for keeping track of the scope of variables.
(B) Symbol table can be implemented using a binary search tree.
(C) Symbol table is not required after the parsing phase.
(D) Symbol table is created during the lexical analysis phase.

2024 CS2 · Q.21 · 1 mark
Consider the following two sets: Set X Set Y P. Lexical Analyzer 1. Abstract Syntax Tree Q. Syntax Analyzer 2. Token R. Intermediate Code Generator 3. Parse Tree S. Code Optimizer 4. Constant Folding Which one of the following options is the CORRECT match from Set X to Set Y ?
(A) P – 4; Q – 1; R – 3; S – 2
(B) P – 2; Q – 3; R – 1; S – 4
(C) P – 2; Q – 1; R – 3; S – 4
(D) P – 4; Q – 3; R – 2; S – 1

2023 · Q.19 · 1 mark
Consider the following definition of a lexical token id for an identifier in a program- ming language, using extended regular expressions: letter → [A-Za-z] digit → [0-9] id → letter (letter | digit)∗ Which one of the following Non-deterministic Finite-state Automata with -transitions accepts the set of valid identifiers? (A double-circle denotes a final state) letter letter digit
(A) letter letter digit
(B) letter letter digit
(C) letter letter digit
(D) CS

2020 · Q.9 · 1 mark [OCR]
Consider the following statements. I. Symbol table is accessed only during lexical analysis and syntax analysis. II. Compilers for programming languages that support recursion necessarily need heap storage for memory allocation in the run-time environment. III. Errors violating the condition ‘any variable must be declared before its use’ are detected during syntax analysis. Which of the above statements is/are TRUE?
(A) Tonly (8) land III only (©) II only (0) None of I, II, and III

2018 · Q.37 · 2 marks
A lexical analyzer uses the following patterns to recognize three tokens T1, T2, and T3 over the alphabet {a,b,c}. T1 : a? (b|c)∗ a T2 : b? (a|c)∗ b T3 : c? (b|a)∗ c Note that ‘x?’ means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix. If the string bbaacabc is processed by the analyzer, which one of the following is the sequence of tokens it outputs?
(A) T1 T2 T3
(B) T1 T1 T3
(C) T2 T1 T3
(D) T3 T3

2017 CS2 · Q.5 · 1 mark [OCR]
Match the following according to input (from the left column) to the compiler phase (in the right column) that processes it: (P) Syntax tree (i) Code generator (Q) Character stream (ii) Syntax analyzer (R) Intermediate representation (iii) Semantic analyzer (S) Token stream (iv) Lexical analyzer
(A) P = (ii). Q > (iii), R > (iv). S > (i)
(B) P= (ii). Q > (i). R— Gi). SS (iv)
(C) P= (iii). Q > (iv). R= (eS > ii)
(D) P > (i). Q > (iv). R"Gii). S > (iii)

B. New practice questions with solutions

1. (1 mark · NAT) A lexical analyzer using the maximal munch rule reads the C fragment a+++b. The number of tokens produced is _____.

4. The tokens are a, ++, + and b.

2. (2 marks · MSQ) Which tasks are performed by the lexical analyzer?

A. Removing comments and white space
B. Detecting unbalanced parentheses
C. Producing tokens together with their lexemes or attributes
D. Checking that operand types match

(A), (C). Balanced parentheses are checked by the parser, and type checking is done in semantic analysis.